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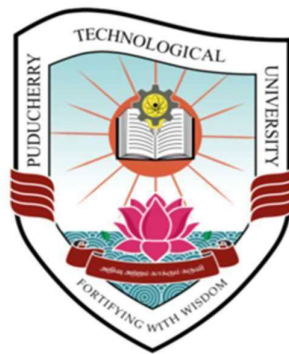
Department of Mechatronics

Project Report Submitted

On

MT222– Engineering Design Project

TITLE: SAFARI: A Vision-Guided Mechatronics Tracking System with
Real-Time Distance Feedback and Mobile Alerting



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ABSTRACT

This project presents the design and implementation of **SAFARI**, a vision-guided mechatronics tracking system developed for applications in wildlife monitoring and forest boundary surveillance. The primary objective of the system is to enable intelligent environmental monitoring by integrating computer vision, embedded control, real-time sensing, and mobile communication into a unified framework. The system is designed as a low-cost, scalable, and semi-autonomous solution capable of detecting, tracking, and responding to dynamic targets in real time.

The vision subsystem is implemented on a laptop using **OpenCV and MediaPipe**, which together provide efficient real-time object detection and tracking capabilities. For demonstration and validation purposes, human hand motion is used as a proxy for wildlife movement. The system processes live video frames, identifies the target region, and continuously computes its spatial position within the camera frame. This positional data is then translated into angular commands, forming the basis for actuator control. The control subsystem is built around an **Arduino UNO** microcontroller, which receives position-based commands via serial communication. These commands are used to drive a **Servo motor**, enabling directional alignment of the system toward the detected target. Unlike discrete step-based control, the system employs continuous angle mapping with smoothing techniques, ensuring fluid, natural, and jitter-free tracking behaviour. This results in a responsive actuation mechanism capable of closely following target motion.

To incorporate environmental awareness, an **ultrasonic distance sensor (HC-SR04)** is mounted directly on the servo mechanism. This configuration ensures that distance measurements are taken along the same line of sight as the tracking direction. **The sensor provides real-time feedback on the proximity of the detected object**, effectively transforming the system into a closed-loop control architecture where sensing and actuation are tightly coupled. A key feature of SAFARI is its communication capability. The system integrates with the **Telegram API to transmit real-time notifications to a mobile device**. Alerts are generated under predefined conditions, such as target detection and critical proximity thresholds. This enables remote monitoring and enhances the system's applicability in safety-critical scenarios such as wildlife intrusion detection, smart fencing, and perimeter security systems.

Overall, the project demonstrates the practical integration of multiple engineering domains, including **computer vision, embedded systems, mechatronics, and IoT-based communication**. SAFARI serves as a proof-of-concept for intelligent surveillance systems that are cost-effective, modular, and adaptable for real-world deployment in environmental monitoring and smart automation applications.

1. INTRODUCTION

With increasing human-wildlife interaction and the need for intelligent monitoring systems, there is a growing demand for low-cost, real-time surveillance solutions in forest environments. Traditional monitoring systems often lack adaptability and real-time response capabilities. This project addresses these limitations by integrating computer vision with embedded systems to create an intelligent tracking and alert mechanism.

The SAFARI system is conceptualized as a **vision-guided mechatronic platform** capable of detecting, tracking, and responding to object movement while providing real-time alerts. The system emphasizes modularity, low cost, and practical deployment potential in wildlife management scenarios.

Furthermore, the SAFARI system is designed with a strong emphasis on cost-effectiveness, scalability, and practical deployment. By using affordable components and open-source technologies, the system ensures accessibility for researchers, forest departments, and small-scale deployments. Overall, this project aims to bridge the gap between advanced surveillance technologies and real-world applicability in wildlife management, contributing to safer and more efficient monitoring solutions

1.1. SYSTEM NAMING AND CONCEPT

The system is named **SAFARI**, inspired by its application in **wildlife observation and forest monitoring environments**. The name reflects the system's ability to observe, track, and respond to movement within natural habitats.

The SAFARI system embodies:

- Intelligent sensing through computer vision
- Automated tracking via servo actuation
- Real-time environmental awareness using distance sensing
- Alert-based response through mobile communication

This naming establishes a direct connection between the system's **technical capabilities** and its **real-world application in forest and wildlife management**.

2. OBJECTIVES

The primary objective of this project is to design and develop an intelligent, vision-based object tracking system that seamlessly integrates computer vision algorithms with embedded control mechanisms. The system aims to bridge the gap between digital perception and physical actuation by enabling real-time mapping of visual input to mechanical motion. Using OpenCV and MediaPipe, the system detects and tracks a target dynamically, while continuously computing its positional coordinates within the camera frame.

These positional parameters are translated into control signals that drive a servo motor through an Arduino UNO microcontroller, enabling precise and continuous orientation toward the detected target. Unlike conventional step-based motion systems, the project emphasizes smooth and adaptive actuation through continuous angle interpolation, ensuring fluid tracking behavior without jerks or discontinuities. This enhances both system responsiveness and mechanical stability.

To extend the system's capabilities beyond visual tracking, ultrasonic sensing is incorporated to measure the real-time distance between the system and the target. The sensor is strategically mounted on the servo mechanism to ensure that distance measurements are aligned with the tracking direction. This integration introduces environmental awareness into the system, allowing it to perceive not only the position but also the proximity of the object.

By combining vision-based detection with distance sensing and actuator control, the system operates as a closed-loop control system. Feedback from the ultrasonic sensor continuously influences system behavior, enabling adaptive responses such as maintaining safe distances or triggering alerts under predefined conditions. This closed-loop architecture improves reliability, accuracy, and overall system intelligence.

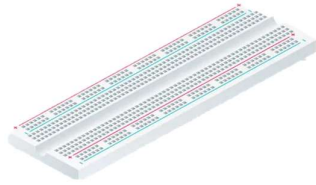
Furthermore, the project incorporates IoT-based communication through integration with the Telegram API, enabling real-time mobile notifications. The system is capable of transmitting alerts when a target is detected or when it enters a critical proximity range. This feature allows remote monitoring and enhances the system's applicability in real-world scenarios where human presence may be limited or delayed.

Overall, the system is designed as a practical, scalable, and cost-effective solution that demonstrates the integration of computer vision, embedded systems, sensor feedback, and wireless communication. The project is particularly relevant for applications such as wildlife monitoring, forest boundary surveillance, smart fencing, and automated environmental observation systems, where real-time tracking and alerting are essential for safety and management.

3. COMPONENTS USED

Hardware

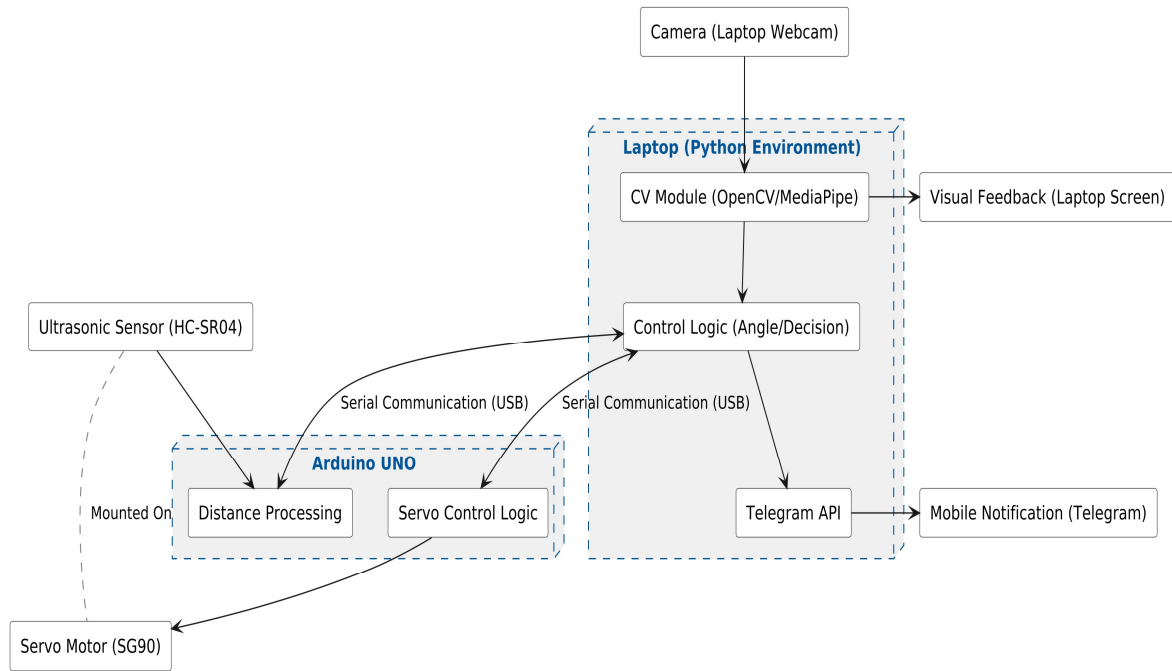
Component	Model / Specification
Microcontroller	Arduino UNO (ATmega328P, 16 MHz) 
Ultrasonic Distance Sensor	HC-SR04 (2–400 cm, ± 3 mm) 
Servo Motor	SG90 (180°, PWM controlled) 
Jumper Wires	Male-to-Male / Male-to-Female Dupont 
USB Cable	USB-A to USB-B (Arduino connection)

Component	Model / Specification
	
Breadboard	400-point mini breadboard 

Software

Component	Model / Specification
Python	v3.10+ for Main scripting language
OpenCV	Real-time video capture and frame processing
MediaPipe	Hand landmark detection and tracking
PySerial	USB serial communication (Python ↔ Arduino)
Requests (Telegram API)	Mobile alert notification via Telegram Bot
Arduino IDE	Embedded firmware development for Arduino UNO

3.1 SYSTEM ARCHITECTURE



The SAFARI system follows a layered architecture integrating multiple domains, including computer vision, embedded systems, sensing, and communication. Each layer performs a specific function, ensuring modularity and scalability of the overall system.

Layers:

1. Input Layer – Camera (Laptop Webcam)

- Captures real-time video frames from the surrounding environment.
- Acts as the primary data acquisition unit for object detection.
- Provides continuous visual input for tracking dynamic targets.
- Ensures sufficient frame rate and resolution for reliable detection.

2. Processing Layer – OpenCV + MediaPipe + Python Logic

- Processes incoming video frames using OpenCV for image handling.
- Utilizes MediaPipe for efficient and real-time hand/object detection.
- Extracts spatial coordinates (position of target in frame).
- Implements decision-making logic for tracking and alert conditions.
- Converts visual data into meaningful control parameters (angle values).

3. Communication Layer – Serial (UART over USB)

- Establishes communication between laptop and Arduino UNO.
- Transmits processed positional data (angle commands) via serial protocol.
- Ensures reliable and low-latency data transfer.
- Acts as a bridge between software (Python) and hardware (microcontroller).

4. Control Layer – Arduino UNO

- Receives serial data and interprets control commands.
- Converts incoming angle values into PWM signals for servo control.
- Executes real-time control logic for smooth actuation.
- Maintains stability and ensures accurate response to input commands.

5. Actuation Layer – Servo Motor

- Performs physical movement based on control signals.
- Rotates smoothly to align with the detected target position.
- Enables directional tracking in real time.
- Provides continuous and responsive motion without abrupt steps.

6. Sensing Layer – Ultrasonic Sensor

- Measures distance between system and detected object.
- Mounted on servo to align sensing direction with tracking direction.
- Provides real-time feedback for environmental awareness.
- Enables detection of proximity thresholds (e.g., too close condition).

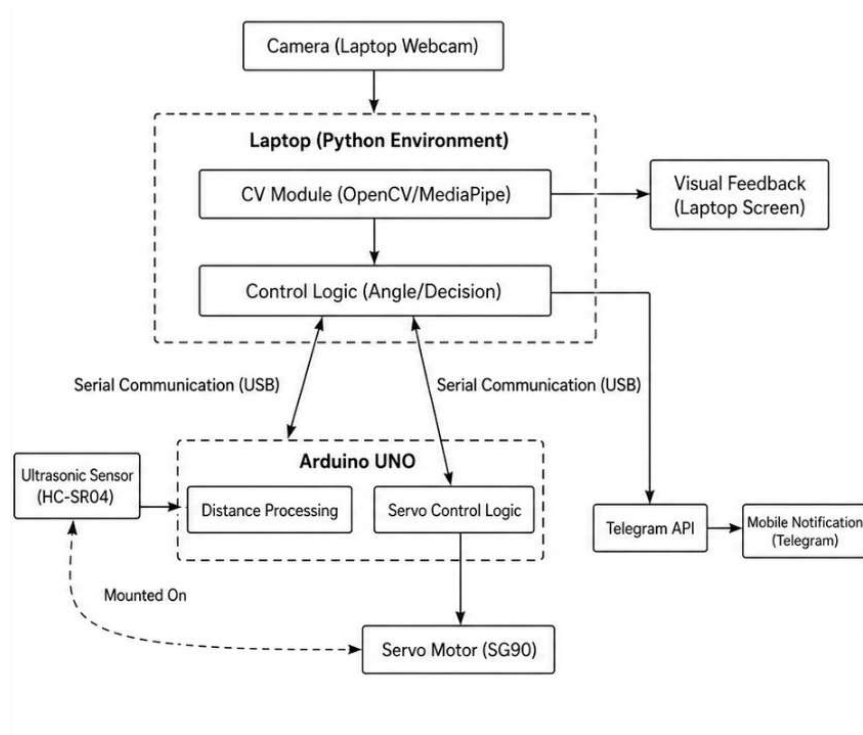
7. Output Layer – Mobile Notification (Telegram API)

- Sends real-time alerts to the user's mobile device.
- Notifies when a target is detected or when it is within critical distance.
- Enables remote monitoring and user awareness.
- Uses internet-based communication for instant notification delivery.

System Behavior

- The system operates as a **closed-loop control system**, where sensor feedback continuously influences system behavior.
- Visual detection drives actuation, while distance sensing provides corrective feedback.
- The integration of feedback ensures adaptive, stable, and intelligent tracking performance.
- This architecture allows the system to respond dynamically to changing environmental conditions.

3.2 HARDWARE ARCHITECTURE



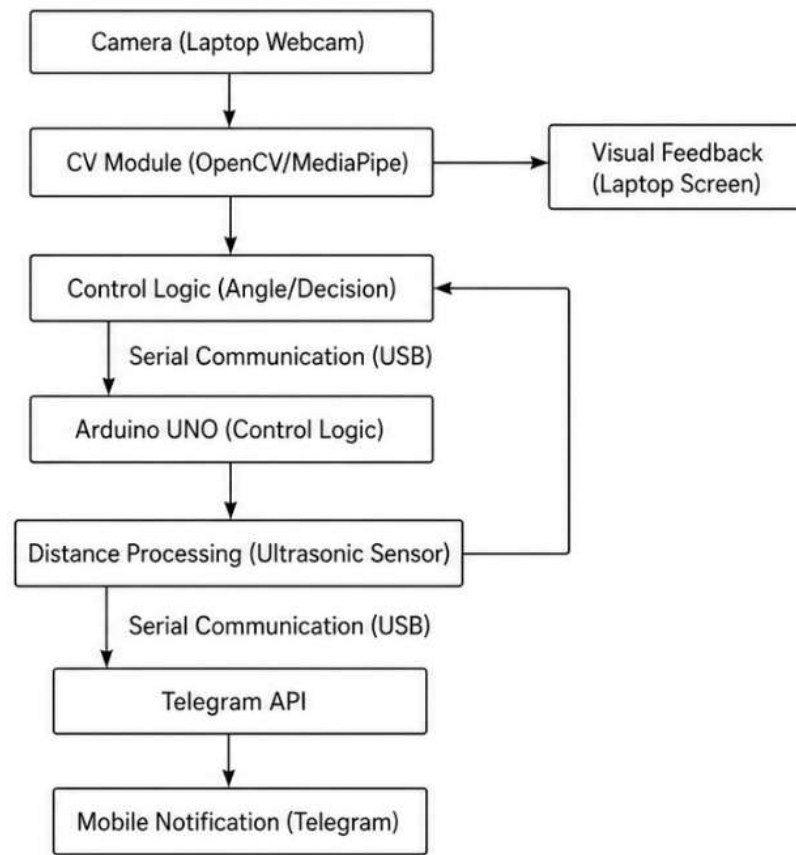
The SAFARI system uses a layered hardware architecture integrating vision processing, embedded control, sensing, and actuation. A laptop webcam captures real-time video, which is processed using OpenCV and MediaPipe to detect the target.

The processed data is sent to the Arduino UNO via serial communication (USB). The Arduino acts as the control unit, driving a servo motor (SG90) for object tracking using PWM signals.

An ultrasonic sensor (HC-SR04) measures the distance to the target and sends feedback to the Arduino. This distance data is transmitted back to the laptop, forming a closed-loop system.

Based on detection and distance conditions, alerts are sent to a mobile device using the Telegram API.

3.3 SOFTWARE ARCHITECTURE



The software architecture follows a sequential pipeline with feedback. Video input is captured and processed using OpenCV, followed by target detection using MediaPipe.

The detected position is mapped to a servo angle and transmitted to the Arduino via serial communication. The Arduino controls the servo motor accordingly.

Distance data from the ultrasonic sensor is sent back to the system, enabling feedback. The decision logic evaluates detection and distance conditions, and notifications are sent through the Telegram API.

4. WORKING PRINCIPLE

1. **The camera captures real-time video input.**
Continuous frames are acquired from the laptop webcam, providing live visual data of the environment. This serves as the primary input for detecting and tracking dynamic targets.
2. **OpenCV and MediaPipe detect and track the hand (target).**
OpenCV processes each frame while MediaPipe identifies and localizes the hand with high efficiency. The system extracts the target's position coordinates within the frame for further processing.
3. **The detected position is converted into a servo angle.**
The horizontal position of the target is mapped proportionally to a corresponding servo angle. This conversion ensures a direct relationship between visual position and actuator movement.
4. **The angle is transmitted to the Arduino via serial communication.**
The computed angle is sent from Python to the Arduino using UART over USB. This communication enables real-time transfer of control commands with minimal delay.
5. **The Arduino drives the servo motor to align with the target.**
The Arduino interprets the received angle and generates PWM signals to control the servo. The servo rotates smoothly to align its direction with the tracked target.
6. **The ultrasonic sensor measures the distance to the object.**
The sensor emits ultrasonic pulses and calculates distance based on echo return time. Since it is mounted on the servo, measurements are aligned with the tracking direction.
7. **Distance data is sent back to the Python environment.**
The Arduino transmits the measured distance values back through serial communication. Python continuously reads and updates this data for monitoring and decision-making.
8. **Python evaluates conditions and triggers alerts via Telegram API.**
The system checks predefined conditions such as target detection and proximity thresholds. If conditions are met, alerts are sent instantly to a mobile device using the Telegram API.

5. MATHEMATICAL MODELING

Mathematical modeling is used in this system to establish a relationship between visual input, actuator response, and distance measurement. It ensures that raw data obtained from sensors and image processing is converted into meaningful control outputs, enabling accurate and real-time system behavior. The model consists of two main components: position-to-angle mapping for actuator control and distance calculation using ultrasonic sensing.

5.1 POSITION TO ANGLE MAPPING

$$\theta = \theta_{min} + (x - x_{min}) \cdot \frac{(\theta_{max} - \theta_{min})}{(x_{max} - x_{min})}$$

This equation represents a linear mapping between the detected position of the object in the camera frame and the corresponding servo motor angle. The variable x denotes the horizontal pixel position of the detected target, while x_{min} and x_{max} define the limits of the camera frame. Similarly, θ_{min} and θ_{max} represent the minimum and maximum rotation angles of the servo motor.

The mapping ensures that the servo motor moves proportionally to the position of the object. When the object is located at the extreme left of the frame, the servo aligns to its minimum angle, and when the object is at the extreme right, it reaches its maximum angle. For intermediate positions, the servo angle is calculated proportionally, resulting in smooth and continuous motion. This eliminates abrupt changes and allows natural tracking behavior.

5.2 DISTANCE CALCULATION

$$d = \frac{t_{echo} \cdot v}{2}$$

The distance between the sensor and the object is calculated using the time-of-flight principle of ultrasonic waves. The sensor emits a sound pulse that travels to the object and reflects back. The total time taken for this round trip is measured as t_{echo} .

Since the signal travels to the object and back, the measured time represents twice the actual distance. Therefore, it is divided by 2 to obtain the true distance. The variable v represents the speed of sound in air, approximately 343 m/s.

By multiplying the time of flight with the speed of sound and dividing by two, the system accurately determines the distance in real time.

6. METHODOLOGY

The methodology adopted for the SAFARI system follows a **systematic, modular, and iterative approach**, integrating computer vision, embedded control, sensing, and communication. The development process is structured into multiple stages, each addressing a specific functional requirement of the system.

6.1 SYSTEM DESIGN APPROACH

The overall design is based on a **closed-loop mechatronic architecture**, where input from the environment is continuously processed and used to control system behavior. The system is divided into two primary domains:

- **High-Level Processing Domain (Laptop):** Responsible for vision processing, decision-making, and communication.
- **Embedded Control Domain (Arduino UNO):** Responsible for actuation and sensor interfacing.

This separation ensures efficient workload distribution and real-time responsiveness.

6.2 VISION-BASED DETECTION AND TRACKING

The first stage of the methodology involves capturing real-time video input using a webcam. Each frame is processed using OpenCV and MediaPipe to detect the presence of a hand, which acts as the target object in this implementation.

MediaPipe identifies multiple landmark points on the hand, from which a bounding region is computed. The central coordinate of this region is extracted and used as a reference for tracking. This approach provides robustness against variations in orientation, scale, and minor environmental disturbances.

The extracted coordinate is then normalized relative to the frame dimensions to ensure consistent mapping to actuator movement.

6.3 COORDINATE MAPPING AND CONTROL SIGNAL GENERATION

The detected position of the target is converted into a corresponding servo angle using a linear mapping function. This ensures that the full range of motion of the servo motor corresponds proportionally to the movement of the target across the camera frame.

To avoid abrupt changes in motion, a smoothing algorithm is applied. Instead of directly assigning the target angle, the system gradually updates the current position using an interpolation factor. This results in continuous and stable motion, improving both tracking accuracy and mechanical reliability.

6.4 SERIAL COMMUNICATION INTERFACE

Once the control signal (servo angle) is generated, it is transmitted to the Arduino UNO using serial communication over USB. The communication protocol is designed to be lightweight and efficient, ensuring minimal latency.

Simultaneously, the Arduino transmits distance measurements obtained from the ultrasonic sensor back to the Python environment. This establishes a bidirectional communication channel, enabling real-time feedback and decision-making.

Proper synchronization and timing mechanisms are implemented to ensure that data transmission remains stable and free from conflicts.

6.5 EMBEDDED CONTROL AND ACTUATION

The Arduino UNO acts as the central embedded controller, receiving angle commands and generating appropriate PWM signals to drive the servo motor. The servo motor is responsible for physically aligning with the detected target, effectively translating visual information into mechanical motion.

The control logic on the Arduino ensures smooth execution of commands while maintaining system stability. Constraints are applied to limit the servo movement within safe operational bounds.

6.6 DISTANCE MEASUREMENT AND FEEDBACK INTEGRATION

An ultrasonic sensor (HC-SR04) is mounted on the servo mechanism to measure the distance between the system and the target. The sensor operates by emitting an ultrasonic pulse and calculating the time taken for the echo to return.

This distance information is continuously monitored and sent back to the Python program. The integration of this sensor creates a **feedback loop**, allowing the system to adapt its behavior based on real-time environmental conditions.

6.7 DECISION LOGIC AND ALERT GENERATION

The Python control system evaluates both the detection status and distance measurements to determine system responses. Two primary conditions are defined:

- **Target Detection Condition:** Triggered when a valid object is identified.
- **Proximity Condition:** Triggered when the measured distance falls below a predefined threshold.

When these conditions are satisfied, the system generates alerts using the Telegram Bot API. A delay-based filtering mechanism is incorporated to prevent redundant notifications and ensure meaningful communication.

6.8 SYSTEM INTEGRATION AND TESTING

After individual modules were developed and validated, they were integrated into a unified system. Extensive testing was conducted under different conditions to evaluate:

- Tracking accuracy
- Response time
- Communication reliability
- Sensor consistency

Iterative refinements were applied to optimize system performance, including tuning the smoothing factor, adjusting detection thresholds, and stabilizing communication.

6.9 WORKFLOW SUMMARY

The overall methodology can be summarized as:

1. Capture video input
2. Detect and track target using vision algorithms
3. Convert position into control signal
4. Transmit command to embedded controller
5. Actuate servo motor
6. Measure distance using ultrasonic sensor
7. Send feedback to processing unit
8. Evaluate conditions and generate alerts

METHODOLOGY OUTCOME

This structured methodology ensures that the SAFARI system operates as a **cohesive, real-time, and intelligent mechatronic solution**, effectively integrating multiple domains into a single functional framework.

7. IMPLEMENTATION DETAILS

7.1 VISION SYSTEM

The vision system forms the core of the tracking mechanism and is implemented using the MediaPipe framework in conjunction with OpenCV. MediaPipe is specifically chosen for its optimized and real-time hand landmark detection capabilities, which are highly efficient even on standard computing hardware. The system processes continuous video frames captured from the laptop camera and identifies key landmarks of the hand, enabling accurate detection of its position.

Once the hand is detected, a bounding region is computed by analyzing the minimum and maximum coordinates of the detected landmarks. From this bounding region, the central position of the hand is determined, which serves as the reference point for tracking. This approach ensures robustness against minor variations in hand shape, orientation, and lighting conditions. The use of real-time computer vision allows the system to dynamically respond to user movement, forming the input for subsequent control and actuation processes.

7.2 CONTROL STRATEGY

To achieve smooth and realistic motion of the servo motor, a continuous control strategy is implemented instead of discrete step-based control. Initially, direct angle commands resulted in abrupt movements, which affected both system stability and tracking accuracy. To overcome this, a smoothing algorithm based on interpolation is applied.

The control equation is given by:

$$current = current + (target - current) \cdot k$$

where *current* represents the present servo position, *target* represents the desired angle, and *k* is a proportional smoothing factor. This approach ensures gradual convergence of the servo towards the target position rather than immediate jumps. As a result, the motion appears fluid and natural, closely resembling real-world tracking behavior. This strategy significantly enhances system responsiveness and reduces mechanical stress on the actuator.

7.3 COMMUNICATION

Communication between the high-level processing unit (laptop) and the embedded system (Arduino UNO) is established using serial communication over USB. The system operates at a baud rate of 9600, ensuring reliable and consistent data transfer without excessive overhead.

The communication protocol is designed to be simple yet effective. The Python program transmits calculated servo angles to the Arduino in real time, enabling continuous actuation. Simultaneously, the Arduino sends back distance measurements obtained from the ultrasonic

sensor in a structured format. This bidirectional communication allows seamless integration between processing and sensing components.

Data flow is maintained as follows:

- Angle values are transmitted from Python to Arduino for actuation.
- Distance values are transmitted from Arduino to Python for feedback and decision-making.

Proper timing control and buffering mechanisms are implemented to prevent data loss and ensure synchronization between both systems.

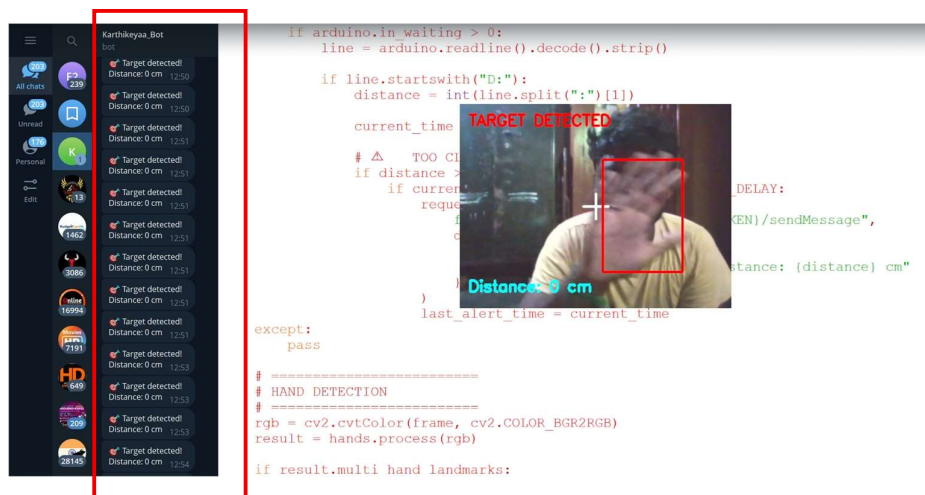
7.4 ALERT SYSTEM

The alert system is implemented using the Telegram Bot API, providing a reliable and scalable method for real-time mobile notifications. This component enables the system to communicate critical information to the user remotely, enhancing its practical usability in monitoring applications.

Two types of alerts are generated:

- Target detection alerts, indicating that an object has been identified and is being tracked.
- Proximity alerts, triggered when the object comes within a predefined distance threshold.

To avoid excessive or redundant notifications, a delay-based control mechanism is incorporated. This ensures that alerts are sent only at controlled intervals, preventing message flooding while maintaining responsiveness. The integration of IoT-based communication transforms the system from a simple tracking setup into an intelligent monitoring solution capable of real-time decision reporting.



8. CHALLENGES FACED

8.1 SERVO INSTABILITY

During the initial stages of implementation, the servo motor exhibited unstable and jerky movements due to step-based control commands. This resulted in poor tracking performance and reduced system smoothness. The issue was addressed by implementing an interpolation-based smoothing algorithm, which allowed gradual transitions between positions. This significantly improved motion stability and tracking accuracy.

8.2 SLOW DETECTION

The system initially experienced delays in object detection due to high computational load from processing high-resolution frames. This affected real-time performance and responsiveness. The problem was resolved by reducing the frame resolution and utilizing a lightweight configuration of the MediaPipe model. These optimizations ensured faster processing while maintaining acceptable detection accuracy.

8.3 SERIAL COMMUNICATION ERRORS

Frequent issues such as port conflicts, buffer overflows, and data misalignment were encountered during serial communication. These problems disrupted synchronization between the Python program and the Arduino. The issue was mitigated by implementing controlled data transmission rates, adding timeout mechanisms, and ensuring that only one process accessed the serial port at a time. This resulted in stable and reliable communication.

8.4 SENSOR ERRORS

The ultrasonic sensor initially produced inconsistent or zero readings due to improper wiring, timing inaccuracies, and environmental interference. These errors were identified through isolated testing of the sensor module. Proper calibration, correct trigger pulse generation, and stable wiring connections resolved the issue, leading to accurate and consistent distance measurements.

8.5 ALERT OVERLOAD

Continuous detection of the target caused repeated triggering of alerts, leading to excessive and redundant notifications. This reduced the usability of the alert system. To address this, a delay-based filtering mechanism was introduced, ensuring that alerts are generated only after a specified time interval. This improved the clarity and effectiveness of notifications.

9. ADVANTAGES

The SAFARI system offers several advantages due to its integrated design and practical implementation:

- It demonstrates seamless **integration of computer vision and embedded systems**, combining software intelligence with hardware control.
- The system operates in **real time**, enabling immediate response to environmental changes.
- It is **cost-effective**, utilizing readily available components and open-source software.
- The modular architecture ensures **scalability**, allowing future enhancements such as multi-axis tracking or cloud integration.
- It provides **practical deployment capability**, particularly in applications such as wildlife monitoring, boundary surveillance, and intrusion detection systems.

Overall, the system represents an efficient and intelligent solution for real-world mechatronics applications.

10. LIMITATIONS

- Dependent on lighting conditions
- Limited to single-axis tracking
- Servo speed constraints
- Camera frame rate limitations

11. APPLICATIONS

- Wildlife monitoring systems
- Forest boundary surveillance
- Smart fencing systems
- Intrusion detection systems
- Human-machine interaction

12. FUTURE SCOPE

- Multi-axis (pan-tilt) tracking
- PID control implementation
- Edge AI deployment
- Cloud-based monitoring dashboard
- Multi-object tracking

13. CONCLUSION

The SAFARI system successfully demonstrates the effective integration of computer vision, embedded control, and IoT-based communication within a unified closed-loop mechatronic framework. By combining real-time object tracking using vision algorithms, precise actuator control through servo mechanisms, and continuous environmental feedback via ultrasonic sensing, the system achieves intelligent and adaptive behavior.

The incorporation of mobile alerting through the Telegram API further enhances the system by enabling remote monitoring and timely decision-making. This multi-layered integration transforms the system from a basic tracking setup into a responsive and intelligent monitoring solution capable of operating in dynamic environments.

From an application perspective, the SAFARI system presents a practical approach to addressing real-world challenges in wildlife monitoring and forest boundary management. It can serve as an early warning mechanism for detecting animal movement near restricted zones or fencing areas, thereby contributing to both wildlife safety and human security.

Furthermore, the project highlights the potential of leveraging low-cost hardware components and open-source software frameworks to develop scalable and efficient surveillance systems. The modular architecture allows for future enhancements such as multi-axis tracking, advanced control strategies, and cloud-based monitoring, making the system adaptable to a wide range of industrial and environmental applications.

In conclusion, SAFARI stands as a robust demonstration of modern mechatronics, where sensing, processing, actuation, and communication are seamlessly integrated to create an intelligent and practical engineering solution.

14. COST ESTIMATION

Component	Qty	Est. Cost (INR)
Arduino UNO (ATmega328P)	1	₹ 550
SG90 Servo Motor (180°)	1	₹ 120
HC-SR04 Ultrasonic Sensor	1	₹ 80
Laptop with Webcam	1	(Pre-existing)
USB-A to USB-B Cable	1	₹ 60
Jumper Wires (M-M and M-F)	1 set	₹ 80
Mini Breadboard (400 point)	1	₹ 60
Miscellaneous (wires, etc.)	1 set	₹ 50
TOTAL		₹ 1,000 Aprox

Approximate estimation — laptop cost not included as it is pre-existing hardware. The total cost of ₹1,000 underscores the cost-effectiveness and accessibility of the SAFARI prototype for educational and research deployment.

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